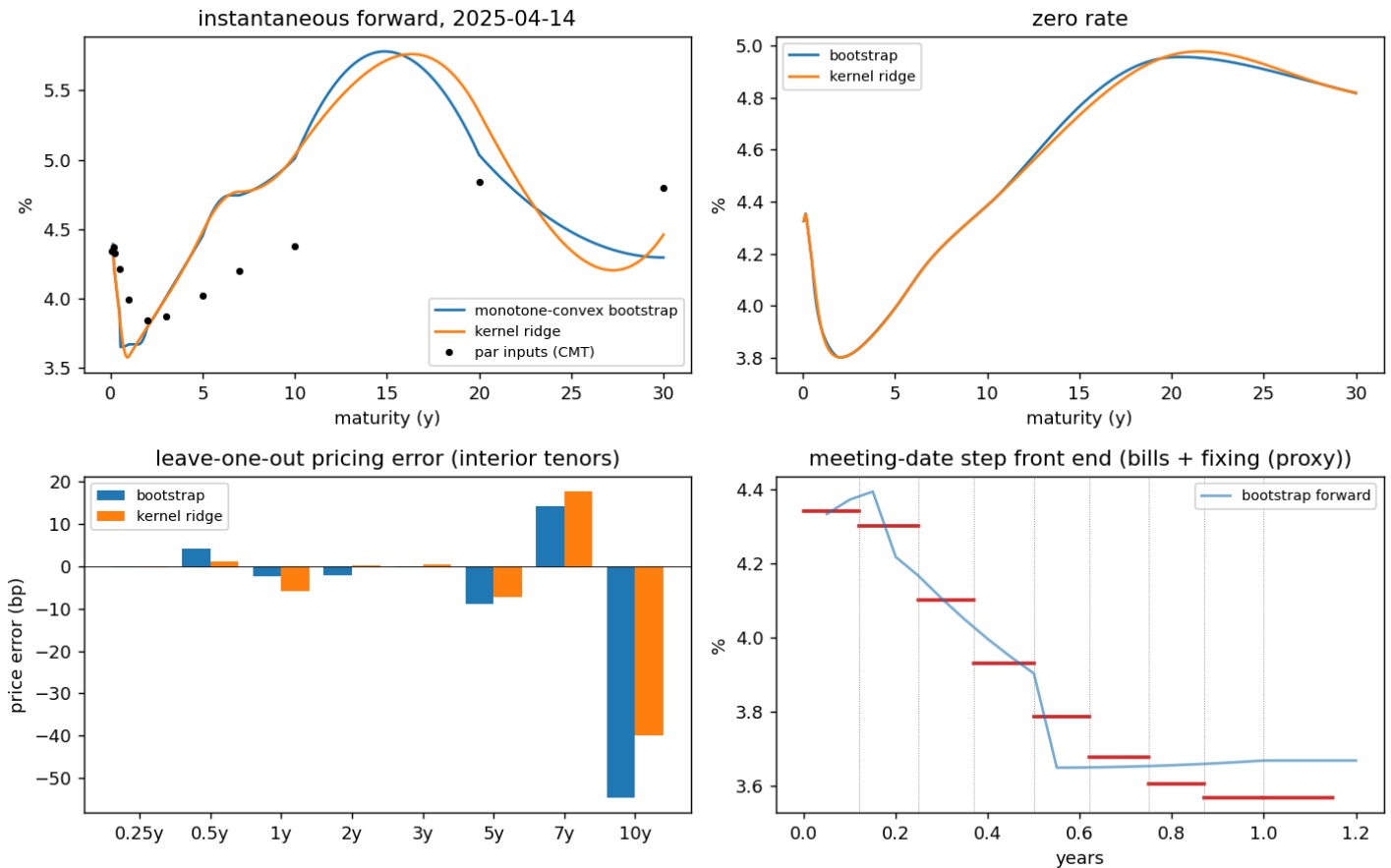


Machine-Learned Yield Curves for Pricing, Hedging and Quoting

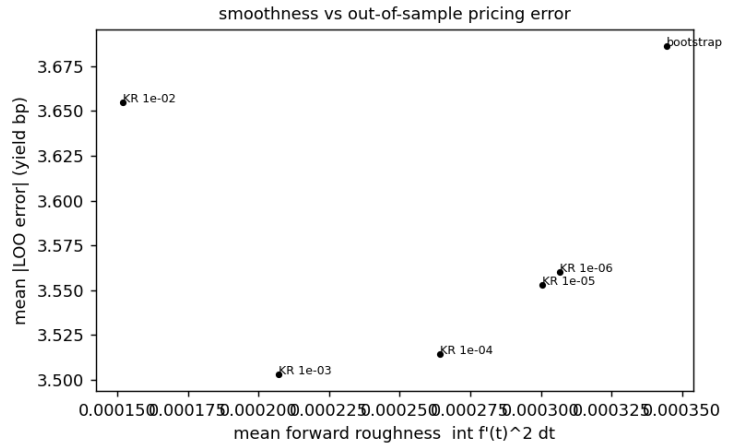
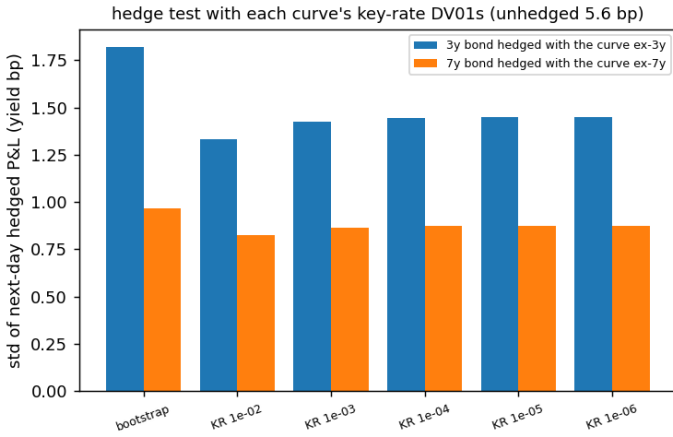
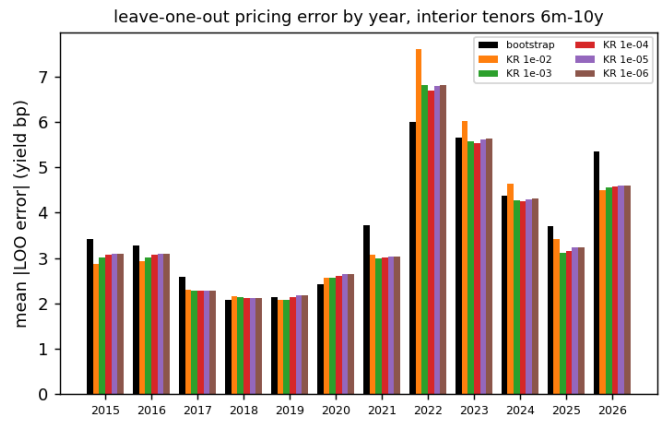
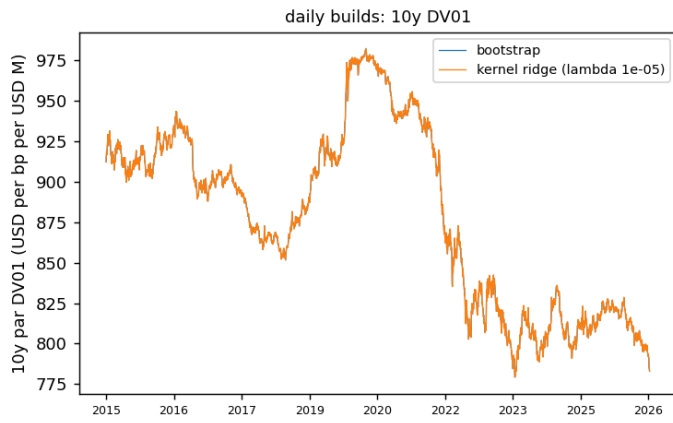
Question. A Treasury RFQ desk needs a curve to price, a curve to hedge and a curve to quote off. Does machine learning improve on the classical answers - a Hagan-West bootstrap, key-rate DV01 hedges, closed-form quotes (Bergault et al., Barzykin-Ciceri, Cartea-Wang) - when each is tested out of sample on 2015-2026 Treasury data and, for quoting, in an RFQ simulator with informed flow and scheduled events?

Method. Header-only C++ behind one CLI. Curve: monotone-convex bootstrap of the Treasury CMT curve with closed-form sector integrals, key-rate DV01s, a meeting-date step front end, and a kernel-ridge discount curve (Filipovic-Pelger-Ye) with the smoothing chosen by a rule fixed in advance. Daily builds 2015-2026 scored by leave-one-out pricing error, forward roughness, DV01 stability and a hedge test (an off-curve bond hedged with the curve's own key-rate DV01s, realised next-day P&L). Hedge ratios learned from co-movements (ridge, PCA) tested walk-forward against the model ratios. Risk: PCA factors; RFQ simulator with tiered clients, informed pre-event flow, event jumps, band hedging, exact P&L decomposition. Quoters: the closed forms and an oracle closed form that knows the client model. Events: FOMC/ECB statement hawkishness (lexicon; FOMC-RoBERTa dated checkpoint), USMPD reactions, NFP/auction calendar, CFTC positioning, funding stress; text signal fitted on 2015-2023 and replayed on 2024-2026 with permutation and look-ahead placebos.

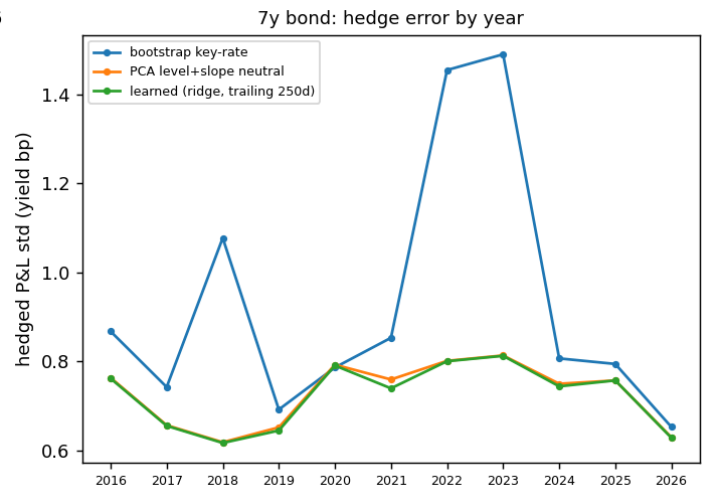
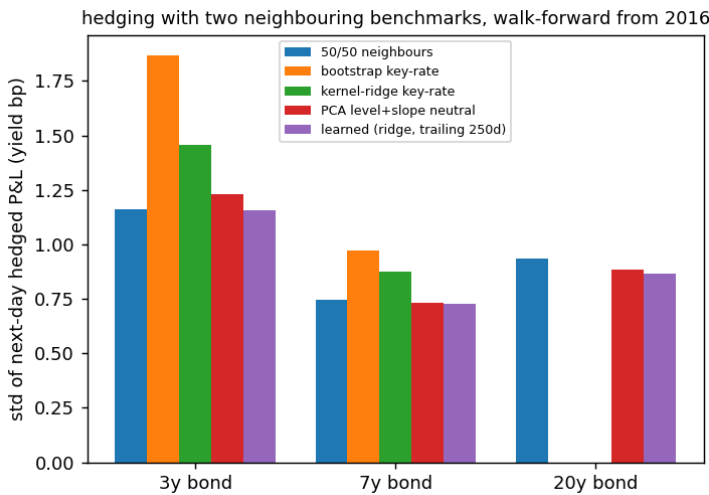
Caveats. SOFR OIS and CME futures are not free: the curve is the Treasury curve and the step front end is fitted from bills. Client flow is simulated. Few meetings per year: text CIs are wide by construction.



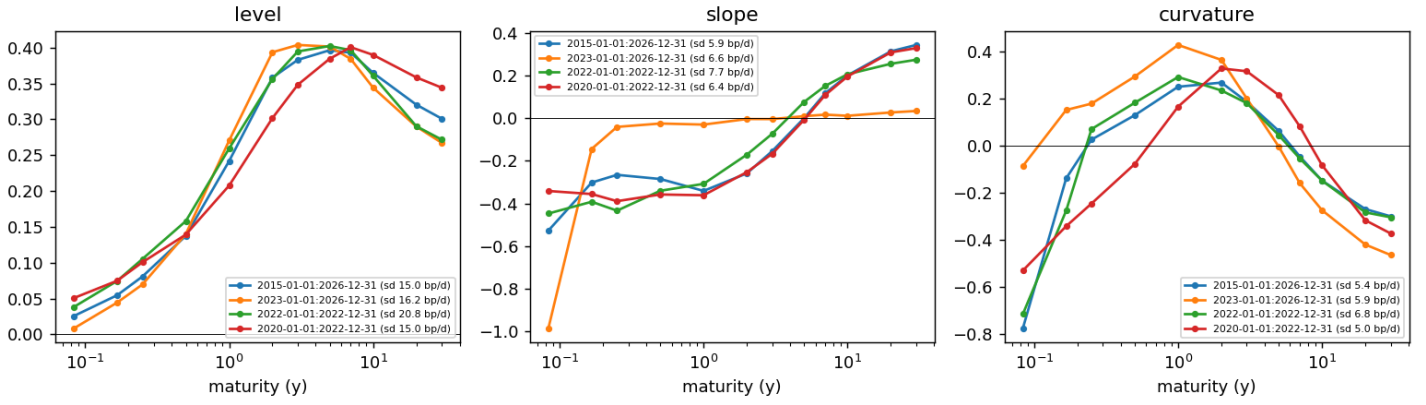
Curve construction on one day: bootstrap vs kernel-ridge forwards and zeros, leave-one-out pricing error (interior tenors), meeting-date step front end.



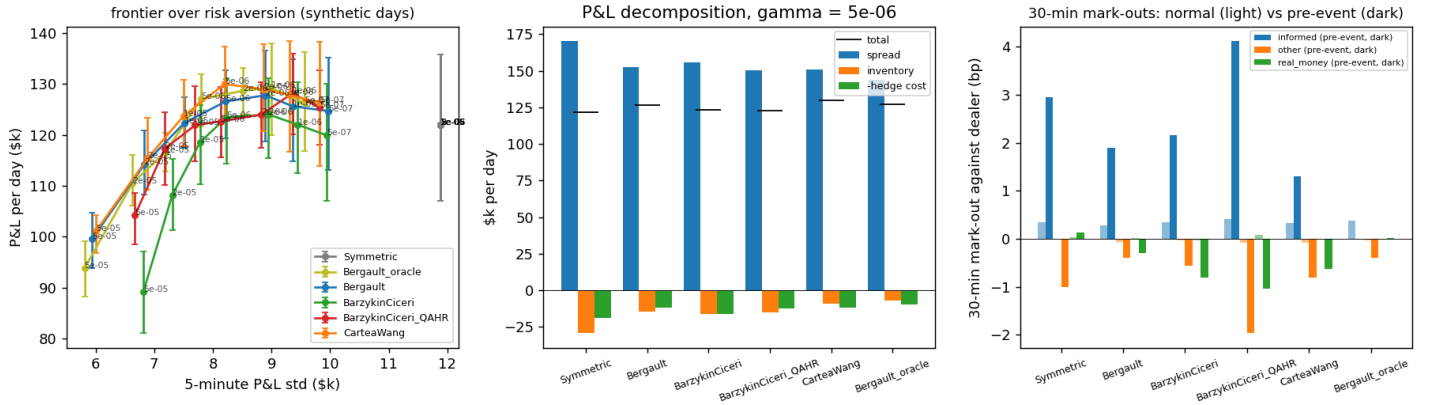
Daily builds 2015-2026: 10y par DV01 under both methods, leave-one-out error by year, hedge test with each curve's key-rate DV01s, smoothness vs error.



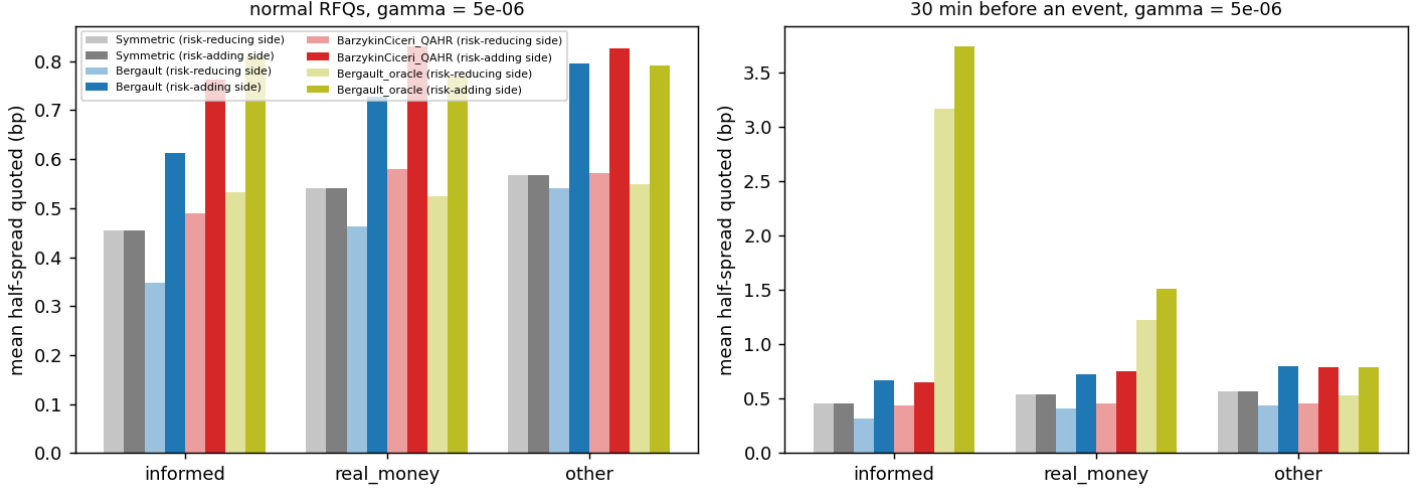
Learned vs model hedge ratios, walk-forward: ridge on the trailing window vs the curves' key-rate DV01s, PCA-neutral and 50/50.



PCA loadings by window: level, slope, curvature and their daily standard deviations.

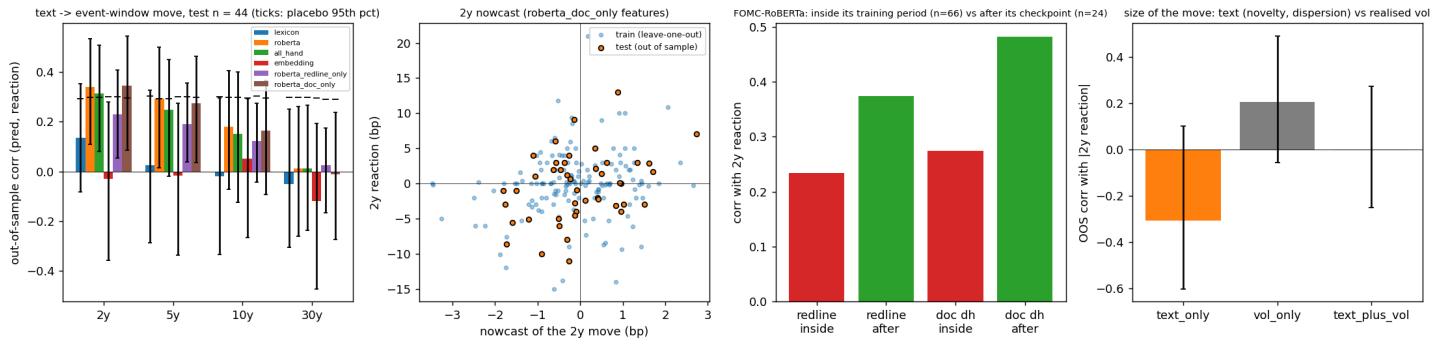


Closed-form quoters on synthetic days: frontier over risk aversion, P&L decomposition, mark-outs by tier around events.

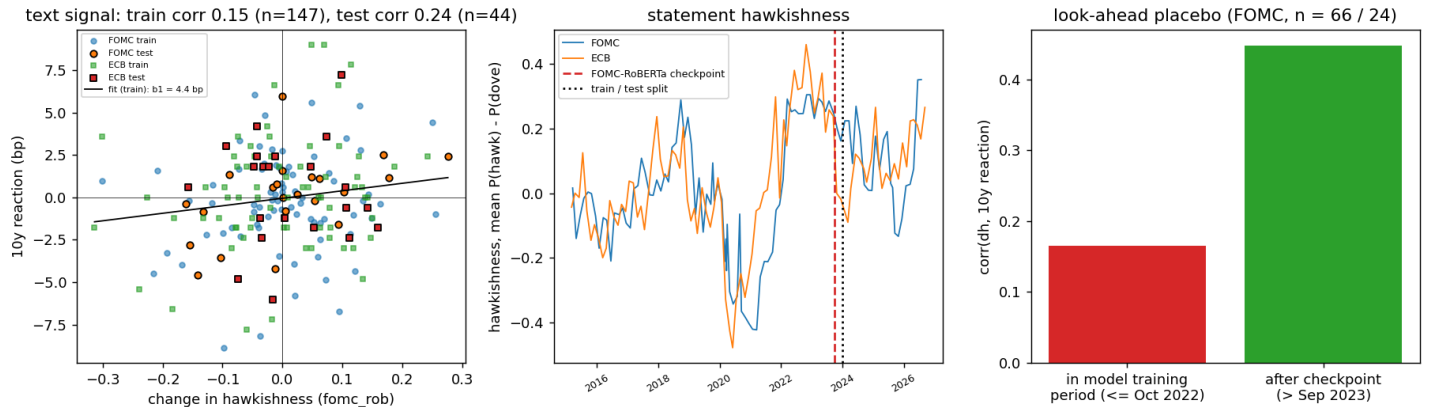


Quote structure of the closed forms: mean half-spread by tier and by risk-reducing / risk-adding side, normal RFQs and the pre-event window.

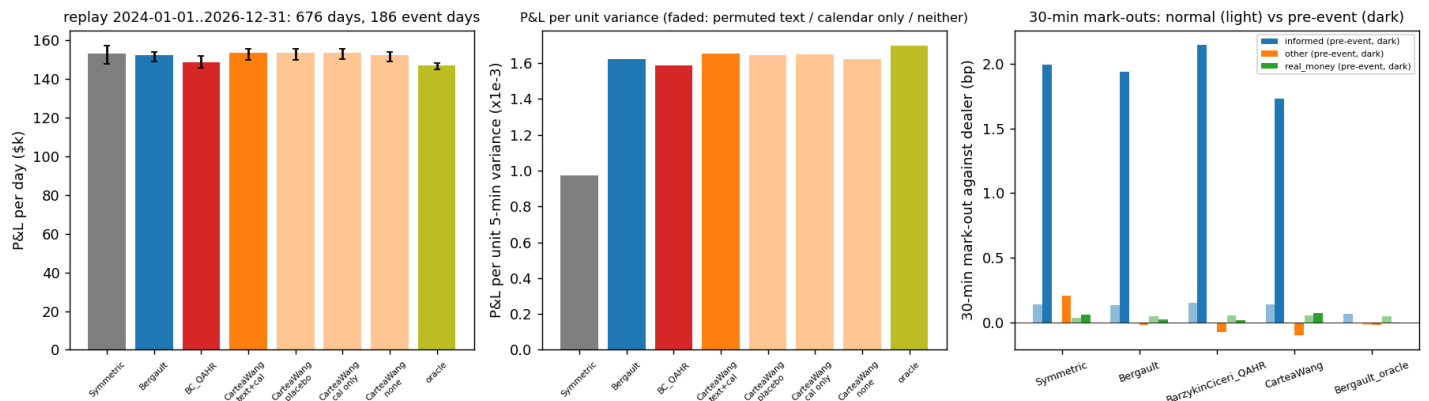
rates-curve-nlp - machine-learned yield curves for pricing, hedging and quoting



NLP arm: out-of-sample correlation of the text nowcast with the event-window move by tenor (ticks = permutation placebo), 2y nowcast scatter, dated-checkpoint split, size-of-move model.



Text signal: change in hawkishness vs 10y reaction (train/test), hawkishness through time, look-ahead placebo.



Real-day replay 2024-2026 with the event features: P&L, P&L per unit variance with signal / permuted signal / no signal, mark-outs around events.

Results summary

Curve construction (2025-04-14)

Bootstrap (Hagan-West monotone convex, global iteration): max repricing error 2.84e-10 bp in 5 sweeps. Kernel ridge ($\lambda = 1e-04$): max repricing error 0.092 bp. Forward roughness $\int f'(t)^2 dt$: bootstrap 2.13e-04, kernel ridge 1.65e-04.

Leave-one-out pricing error, interior tenors 6m-10y (mean |err|): bootstrap 10.85 bp, kernel ridge 9.08 bp. Endpoints are extrapolation and excluded (20y: 224 / 347 bp; 30y: -361 / -1511 bp).

				tenor
0.0833333y	4.340	-0.48	-0.40	
0.166667y	4.370	+0.47	+0.43	
0.25y	4.330	-0.24	-0.36	
0.5y	4.210	+4.15	+1.05	
1y	3.990	-2.36	-5.87	
2y	3.840	-2.14	+0.08	
3y	3.870	-0.35	+0.46	
5y	4.020	-8.81	-7.28	
7y	4.200	+14.18	+17.61	
10y	4.380	-54.56	-39.92	
20y	4.840	+224.49	+347.12	
30y	4.800	-360.87	-1511.17	

Ladder key-rate DV01s (per 1 notional per bp; row = bond, columns = bumped input):

														bond
2y	3.840	1.91e-04	-2.2e-16	-2.2e-16	-2.2e-16	2.9e-15	0.0e+00	1.9e-04	-8.9e-16	-2.2e-16	0.0e+00	0.0e+00	0.0e+00	0.0e+00
5y	4.020	4.50e-04	0.0e+00	1.1e-16	0.0e+00	1.1e-16	0.0e+00	0.0e+00	-1.3e-15	4.5e-04	-3.1e-14	1.1e-14	-6.7e-16	-4.4e-16
10y	4.380	8.10e-04	0.0e+00	-2.2e-16	-2.2e-16	0.0e+00	0.0e+00	-2.2e-16	0.0e+00	0.0e+00	0.0e+00	8.1e-04	0.0e+00	-2.2e-16
30y	4.800	1.59e-03	2.2e-16	0.0e+00	0.0e+00	2.2e-16	5.6e-16	0.0e+00	0.0e+00	4.4e-16	4.4e-16	-2.2e-16	0.0e+00	1.6e-03

Meeting-date step front end (bills + fixing (proxy)): levels 4.34%, 4.30%, 4.10%, 3.93%, 3.79%, 3.68%, 3.61%, 3.57%, 3.57% on 0.12y, 0.25y, 0.37y, 0.50y, 0.62y, 0.75y, 0.87y, 1.00y.

Daily curve builds, 2015-01-02 .. 2026-09-15 (2927 days)

Selection rule fixed in advance: the smoothest kernel ridge (largest λ) whose max repricing error over all days is within 1 bp; that is $\lambda = 1e-05$. Leave-one-out = each interior tenor (6m-10y) dropped and repriced off the curve built from the rest; hedge test = a 3y (7y) par bond priced off the curve built without that point and hedged with the remaining inputs using that curve's key-rate DV01s, realised next-day hedged P&L in yield bp.

									method
bootstrap	-	2.8e-07	3.686	3.457	2.248	3.44e-04	1.822	0.965	
kernel_ridge	1e-02	10	3.655	3.089	2.251	1.52e-04	1.331	0.827	
kernel_ridge	1e-03	5.3	3.503	2.996	2.252	2.07e-04	1.424	0.866	
kernel_ridge	1e-04	2.2	3.514	3.066	2.252	2.64e-04	1.446	0.872	
kernel_ridge	1e-05	0.32	3.553	3.097	2.252	3.00e-04	1.451	0.873	
kernel_ridge	1e-06	0.034	3.560	3.105	2.252	3.07e-04	1.451	0.873	

Selected kernel ridge vs bootstrap: LOO error -3.6% (median -10.4%), roughness -12.8%, hedged 3y / 7y std -20.4% / -9.6%, 10y DV01 day-to-day std +0.2%. Unhedged 7y daily std 5.55 bp.

												year
bootstrap	3.42	3.27	2.59	2.08	2.13	2.43	3.71	6.01	5.66	4.38	3.70	5.36
KR 1e-02	2.87	2.92	2.30	2.15	2.07	2.57	3.06	7.61	6.02	4.65	3.41	4.50
KR 1e-03	3.00	3.00	2.27	2.12	2.07	2.57	2.99	6.82	5.57	4.27	3.10	4.57
KR 1e-04	3.08	3.07	2.27	2.11	2.14	2.61	3.01	6.70	5.53	4.26	3.15	4.57
KR 1e-05	3.09	3.08	2.27	2.12	2.18	2.64	3.03	6.80	5.62	4.30	3.23	4.60
KR 1e-06	3.09	3.09	2.27	2.12	2.18	2.64	3.04	6.82	5.64	4.31	3.24	4.61

Learned vs model hedge ratios (walk-forward from 2016-01-04, trailing window 250 days)

bond	hedge ratios	days	hedged P&L std (yield bp)	reduction vs unhedged
3y	unhedged	2675	5.473	0.0%
3y	50/50 neighbours	2675	1.159	78.8%
3y	bootstrap key-rate DV01s	2675	1.868	65.9%
3y	kernel-ridge key-rate DV01s	2675	1.456	73.4%

3y	PCA level+slope neutral (trailing window 100 days)	2675	1.230	77.5%
3y	ridge regression on neighbours (trailing window 100 days)	2675	1.158	78.8%
7y	unhedged	2675	5.555	0.0%
7y	50/50 neighbours	2675	0.745	86.6%
7y	bootstrap key-rate DV01s	2675	0.974	82.5%
7y	kernel-ridge key-rate DV01s	2675	0.873	84.3%
7y	PCA level+slope neutral (trailing window 100 days)	2675	0.731	86.8%
7y	ridge regression on neighbours (trailing window 100 days)	2675	0.728	86.9%
20y	unhedged	2676	5.084	0.0%
20y	50/50 neighbours	2676	0.935	81.6%
20y	PCA level+slope neutral (trailing window 100 days)	2676	0.885	82.6%
20y	ridge regression on neighbours (trailing window 100 days)	2676	0.868	82.9%

3y: learned / bootstrap-key-rate std ratio 0.620 (bootstrap 95% CI [0.552, 0.692]).

7y: learned / bootstrap-key-rate std ratio 0.747 (bootstrap 95% CI [0.699, 0.793]).

Factor model (PCA of daily par-yield changes, 12 tenors)

window	days	explained 1/2/3	factor sd bp/day (level, slope, curvature)
2015-01-01:2026-12-31	1977	0.708 / 0.817 / 0.908	15.0, 5.9, 5.4
2023-01-01:2026-12-31	926	0.713 / 0.830 / 0.924	16.2, 6.6, 5.9
2022-01-01:2022-12-31	249	0.723 / 0.823 / 0.902	20.8, 7.7, 6.8
2020-01-01:2022-12-31	751	0.702 / 0.831 / 0.908	15.0, 6.4, 5.0

Closed-form quoters on synthetic days (40 days x 8 seeds, common random numbers)

P&L per day with 95% bootstrap CI over seeds; 5-minute P&L std; decomposition (spread + inventory - hedge cost = total, residual is machine zero); hit ratios by tier (informed / real money / other).

quoter

Symmetric	5e-07	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Symmetric	1e-06	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Symmetric	2e-06	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Symmetric	5e-06	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Symmetric	1e-05	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Symmetric	2e-05	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Symmetric	5e-05	122.0	[107.1, 135.8]	11.9	0.86	170.4	-29.3	-19.1	0.0e+00	0.34/0.36/0.35
Bergault	5e-07	124.7	[113.1, 135.2]	10.0	1.26	166.6	-25.9	-16.0	0.0e+00	0.37/0.35/0.32
Bergault	1e-06	125.6	[114.8, 134.9]	9.4	1.43	163.7	-23.2	-14.9	0.0e+00	0.36/0.35/0.32
Bergault	2e-06	127.8	[118.7, 136.6]	8.9	1.61	161.5	-19.8	-13.9	0.0e+00	0.35/0.35/0.31
Bergault	5e-06	126.6	[119.4, 132.7]	8.2	1.87	152.8	-14.5	-11.7	0.0e+00	0.34/0.33/0.30
Bergault	1e-05	122.3	[117.5, 127.5]	7.5	2.17	144.8	-12.7	-9.8	0.0e+00	0.33/0.32/0.29
Bergault	2e-05	114.2	[108.4, 120.8]	6.8	2.45	131.8	-9.6	-8.0	0.0e+00	0.30/0.30/0.28
Bergault	5e-05	99.5	[93.8, 104.7]	5.9	2.82	111.4	-6.4	-5.5	0.0e+00	0.26/0.27/0.26
BarzykinCiceri	5e-07	119.9	[107.1, 130.1]	9.9	1.21	165.4	-27.7	-17.8	0.0e+00	0.35/0.36/0.35
BarzykinCiceri	1e-06	122.0	[112.5, 130.4]	9.4	1.37	163.8	-24.4	-17.4	0.0e+00	0.35/0.36/0.35
BarzykinCiceri	2e-06	124.0	[115.4, 131.1]	8.9	1.55	161.5	-20.5	-16.9	0.0e+00	0.35/0.36/0.35
BarzykinCiceri	5e-06	123.4	[114.3, 131.0]	8.2	1.82	155.6	-16.0	-16.2	0.0e+00	0.36/0.36/0.36
BarzykinCiceri	1e-05	118.6	[110.3, 125.8]	7.8	1.96	148.3	-14.0	-15.6	0.0e+00	0.36/0.36/0.36
BarzykinCiceri	2e-05	108.1	[101.3, 115.3]	7.3	2.03	136.8	-13.7	-14.9	0.0e+00	0.36/0.36/0.36
BarzykinCiceri	5e-05	89.2	[81.0, 97.2]	6.8	1.92	118.5	-14.8	-14.5	0.0e+00	0.37/0.37/0.36
BarzykinCiceri	5e-06RR	125.4	[118.1, 132.7]	9.8	1.30	158.6	-19.2	-14.0	0.0e+00	0.26/0.28/0.28
BarzykinCiceri	1e-06RR	128.2	[120.1, 136.0]	9.4	1.46	157.1	-15.6	-13.4	0.0e+00	0.26/0.28/0.28
BarzykinCiceri	2e-06RR	124.0	[117.5, 130.4]	8.8	1.59	155.8	-18.7	-13.2	0.0e+00	0.26/0.28/0.28
BarzykinCiceri	5e-06RR	122.7	[115.7, 129.5]	8.1	1.86	150.5	-15.3	-12.6	0.0e+00	0.26/0.28/0.28
BarzykinCiceri	1e-05RR	121.9	[114.8, 129.7]	7.7	2.06	146.2	-12.2	-12.1	0.0e+00	0.26/0.28/0.28
BarzykinCiceri	2e-05RR	117.3	[110.2, 124.5]	7.2	2.28	138.2	-9.2	-11.7	0.0e+00	0.26/0.28/0.28
BarzykinCiceri	5e-05RR	104.3	[98.4, 108.6]	6.7	2.34	125.4	-10.0	-11.1	0.0e+00	0.27/0.28/0.29
CarteaWang	5e-07	126.3	[113.9, 138.3]	9.8	1.31	164.0	-21.7	-16.0	0.0e+00	0.35/0.35/0.32
CarteaWang	1e-06	127.8	[116.6, 138.5]	9.3	1.48	161.4	-18.6	-15.0	0.0e+00	0.35/0.35/0.32
CarteaWang	2e-06	129.0	[120.7, 137.8]	8.9	1.64	158.8	-15.9	-13.8	0.0e+00	0.34/0.34/0.31

rates-curve-nlp - machine-learned yield curves for pricing, hedging and quoting

CarteaWang	5e-06	130.0	[123.7, 137.4]	8.2	1.93	150.7	-9.0	-11.8	0.0e+00	0.32/0.33/0.30
CarteaWang	1e-05	123.7	[117.7, 130.9]	7.5	2.20	141.8	-8.2	-9.9	0.0e+00	0.31/0.32/0.29
CarteaWang	2e-05	115.4	[109.2, 123.3]	6.9	2.43	129.3	-6.0	-7.9	0.0e+00	0.29/0.30/0.28
CarteaWang	5e-05	101.1	[96.8, 104.3]	6.0	2.80	110.2	-3.6	-5.5	0.0e+00	0.26/0.27/0.26
Bergault_oracle	5e-07	126.3	[116.9, 136.3]	9.6	1.38	155.9	-16.4	-13.3	0.0e+00	0.22/0.31/0.32
Bergault_oracle	1e-06	129.2	[119.9, 138.0]	9.0	1.59	153.9	-12.3	-12.4	0.0e+00	0.21/0.31/0.32
Bergault_oracle	2e-06	128.7	[124.0, 133.2]	8.5	1.77	150.2	-10.1	-11.5	0.0e+00	0.21/0.30/0.31
Bergault_oracle	5e-06	127.0	[121.6, 131.8]	7.8	2.09	143.7	-6.9	-9.8	0.0e+00	0.21/0.29/0.30
Bergault_oracle	1e-05	116.6	[112.8, 120.5]	7.2	2.26	134.4	-9.5	-8.3	0.0e+00	0.20/0.28/0.29
Bergault_oracle	2e-05	110.5	[106.1, 116.1]	6.6	2.51	123.9	-6.6	-6.8	0.0e+00	0.19/0.26/0.28
Bergault_oracle	5e-05	93.8	[88.3, 99.2]	5.8	2.78	103.9	-5.5	-4.6	0.0e+00	0.18/0.24/0.25

Mark-outs at 30 minutes (bp against the dealer), gamma = 5e-06:

								quoter
Symmetric	informed	normal	278	0.45	+0.35	+0.34	+0.31	
Symmetric	informed	pre-event	14	0.45	+2.94	+2.94	+3.10	
Symmetric	real_money	normal	575	0.54	+0.04	+0.03	+0.02	
Symmetric	real_money	pre-event	39	0.54	+0.14	+0.13	+0.21	
Symmetric	other	normal	814	0.57	+0.00	+0.00	+0.03	
Symmetric	other	pre-event	16	0.57	-1.01	-1.01	-0.90	
Bergault	informed	normal	270	0.40	+0.28	+0.28	+0.30	
Bergault	informed	pre-event	11	0.37	+1.90	+1.90	+2.19	
Bergault	real_money	normal	519	0.53	+0.01	+0.01	+0.04	
Bergault	real_money	pre-event	39	0.50	-0.29	-0.29	-0.31	
Bergault	other	normal	700	0.61	-0.06	-0.06	-0.04	
Bergault	other	pre-event	14	0.55	-0.50	-0.39	-0.33	
BarzykinCiceri	informed	normal	282	0.39	+0.35	+0.34	+0.28	
BarzykinCiceri	informed	pre-event	15	0.38	+2.15	+2.15	+2.32	
BarzykinCiceri	real_money	normal	569	0.49	-0.04	-0.04	-0.03	
BarzykinCiceri	real_money	pre-event	49	0.40	-0.81	-0.81	-0.80	
BarzykinCiceri	other	normal	830	0.48	-0.04	-0.04	-0.03	
BarzykinCiceri	other	pre-event	17	0.49	-0.59	-0.57	-0.42	
BarzykinCiceri_QAHR	informed	normal	198	0.52	+0.42	+0.41	+0.35	
BarzykinCiceri_QAHR	informed	pre-event	13	0.41	+4.12	+4.12	+4.47	
BarzykinCiceri_QAHR	real_money	normal	429	0.62	+0.09	+0.09	-0.00	
BarzykinCiceri_QAHR	real_money	pre-event	42	0.47	-1.04	-1.03	-1.01	
BarzykinCiceri_QAHR	other	normal	666	0.61	-0.08	-0.08	-0.07	
BarzykinCiceri_QAHR	other	pre-event	13	0.57	-1.97	-1.97	-1.83	
CarteaWang	informed	normal	277	0.39	+0.33	+0.33	+0.34	
CarteaWang	informed	pre-event	4	0.80	+1.30	+1.30	+1.47	
CarteaWang	real_money	normal	518	0.52	-0.03	-0.02	-0.02	
CarteaWang	real_money	pre-event	31	0.62	-0.63	-0.62	-0.58	
CarteaWang	other	normal	715	0.59	-0.09	-0.09	-0.06	
CarteaWang	other	pre-event	9	0.70	-0.80	-0.80	-0.89	
Bergault_oracle	informed	normal	190	0.53	+0.39	+0.39	+0.36	
Bergault_oracle	informed	pre-event	0	0.00	+0.00	+0.00	+0.00	
Bergault_oracle	real_money	normal	469	0.57	-0.02	-0.01	-0.02	
Bergault_oracle	real_money	pre-event	8	1.29	+0.18	+0.02	+0.10	
Bergault_oracle	other	normal	699	0.61	-0.04	-0.03	-0.04	
Bergault_oracle	other	pre-event	14	0.55	-0.52	-0.40	-0.48	

Text signal (fomc_rob), training window to 2023-12-31

Regression of the 10y reaction on the change in statement hawkishness, pooled FOMC + ECB: b1 = 4.4 bp per unit (bootstrap 95% CI [-0.1, 9.0]), n = 147, in-sample corr 0.150, R2 0.023. Out of sample (n = 44): corr 0.2407691934597632, R2 0.06749162310294932, sign hit rate 0.5227272727272727.

By bank (training window): FOMC: n 76, corr(dh) 0.125, corr(level) 0.117; ECB: n 71, corr(dh) 0.174, corr(level) -0.088.

Look-ahead placebo (FOMC-RoBERTa checkpoint 2023-09-26): corr(dh, reaction) inside the model's training period 0.1647988045987716 (n = 66) vs after the checkpoint 0.4481200049629283 (n = 24).

NLP: statement text -> event-window move (train to 2023-12-31, n = 147; test n = 44)

Ridge on hand-scored features (lexicon and FOMC-RoBERTa, whole document and redline = sentences added minus removed vs the previous statement, novelty, dispersion) or on the 384-d MiniLM embedding of the redline; lambda by leave-one-out on the training set; placebo = permutation test of the out-of-sample correlation (95th percentile of |corr| under permutation, and the two-sided p-value).

									features
lexicon	2y	100	-0.33	0.13 [-0.08, 0.35]	0.008	0.50 [0.34, 0.64]	0.29	0.392	
lexicon	5y	10	-0.28	0.02 [-0.29, 0.33]	-0.007	0.57 [0.43, 0.70]	0.30	0.871	
lexicon	10y	10	-0.12	-0.02 [-0.34, 0.30]	-0.025	0.50 [0.34, 0.64]	0.30	0.898	
lexicon	30y	1	-0.00	-0.05 [-0.31, 0.25]	-0.061	0.50 [0.36, 0.64]	0.30	0.750	
roberta	2y	100	0.07	0.34 [0.11, 0.53]	0.075	0.48 [0.32, 0.64]	0.30	0.023	
roberta	5y	0.1	0.08	0.29 [0.02, 0.50]	0.076	0.57 [0.43, 0.70]	0.29	0.052	
roberta	10y	0.1	0.00	0.18 [-0.07, 0.41]	0.015	0.48 [0.34, 0.61]	0.30	0.235	
roberta	30y	30	-0.12	0.01 [-0.26, 0.26]	-0.036	0.45 [0.32, 0.61]	0.30	0.941	
all_hand	2y	100	0.00	0.31 [0.08, 0.51]	0.070	0.48 [0.34, 0.64]	0.30	0.037	
all_hand	5y	30	0.02	0.25 [-0.02, 0.45]	0.060	0.59 [0.45, 0.73]	0.29	0.100	
all_hand	10y	10	-0.02	0.15 [-0.12, 0.40]	0.011	0.50 [0.36, 0.64]	0.30	0.325	
all_hand	30y	30	-0.12	0.01 [-0.24, 0.26]	-0.039	0.43 [0.30, 0.57]	0.30	0.935	
embedding	2y	1	0.09	-0.03 [-0.36, 0.28]	-4.243	0.52 [0.39, 0.66]	0.30	0.843	
embedding	5y	1	-0.06	-0.02 [-0.34, 0.27]	-7.494	0.55 [0.39, 0.68]	0.30	0.905	
embedding	10y	300	-0.12	0.05 [-0.26, 0.29]	-0.223	0.57 [0.41, 0.70]	0.29	0.737	
embedding	30y	1000	-0.14	-0.12 [-0.47, 0.19]	-0.147	0.45 [0.30, 0.59]	0.29	0.440	
roberta_redline_only	2y	0.1	0.12	0.23 [0.05, 0.41]	0.040	0.55 [0.39, 0.70]	0.30	0.131	
roberta_redline_only	5y	0.1	0.11	0.19 [0.04, 0.36]	0.031	0.55 [0.39, 0.70]	0.30	0.209	
roberta_redline_only	10y	0.1	0.03	0.12 [-0.04, 0.27]	0.011	0.50 [0.34, 0.64]	0.30	0.415	
roberta_redline_only	30y	0.1	-0.37	0.03 [-0.17, 0.18]	-0.004	0.57 [0.43, 0.73]	0.29	0.866	
roberta_doc_only	2y	0.1	0.17	0.34 [0.09, 0.54]	0.091	0.57 [0.43, 0.73]	0.29	0.022	
roberta_doc_only	5y	0.1	0.17	0.27 [0.03, 0.46]	0.074	0.57 [0.43, 0.73]	0.30	0.068	
roberta_doc_only	10y	0.1	0.07	0.16 [-0.09, 0.37]	0.025	0.52 [0.39, 0.68]	0.29	0.282	
roberta_doc_only	30y	0.1	-0.53	-0.01 [-0.27, 0.24]	-0.005	0.59 [0.45, 0.73]	0.29	0.947	

Dated-checkpoint arm (FOMC-RoBERTa checkpoint 2023-09-26, training data to 2022-10): corr(redline score, 2y reaction) inside the training period 0.23 (n = 66) vs after the checkpoint 0.37 (n = 24); document-level dh 0.27 vs 0.48.

Size of the move (|2y reaction|), OOS corr: text_only: -0.31 [-0.60, 0.10]; vol_only: 0.21 [-0.06, 0.49]; text_plus_vol: -0.00 [-0.25, 0.27]. Replay alphas come from the roberta_doc_only feature set.

Real-day replay with event features: 2024-01-01..2026-12-31, 676 days, 186 event days (21 with measured US

Events by type: AUCTION 111, ECB 22, FOMC 22, NFP 31.

										quoter
Symmetric	text+calendar	152.8	[147.7, 157.2]	12.5	0.97	185.4	-10.1	-22.4	0.35/0.35/0.35	
Bergault	text+calendar	152.1	[149.1, 153.8]	9.7	1.62	173.5	-6.2	-15.1	0.35/0.33/0.31	
BarzykinCiceri_QADR	text+calendar	148.3	[145.8, 151.9]	9.7	1.59	169.4	-6.0	-15.1	0.26/0.27/0.28	
CarteaWang	text+calendar	153.4	[149.9, 155.5]	9.6	1.65	172.3	-4.0	-14.9	0.34/0.33/0.31	
CarteaWang	placebo_text+calendar	153.5	[149.7, 155.6]	9.7	1.64	172.2	-3.8	-14.9	0.34/0.33/0.31	
CarteaWang	calendar_only	153.6	[150.3, 155.4]	9.6	1.65	172.4	-3.9	-14.9	0.34/0.33/0.31	
CarteaWang	none	152.1	[149.1, 153.8]	9.7	1.62	173.5	-6.2	-15.1	0.35/0.33/0.31	
Bergault_oracle	text+calendar	146.9	[144.7, 148.2]	9.3	1.70	160.6	-1.3	-12.4	0.21/0.29/0.31	

quoter	tier	window	fills	half-spread bp	30m mark-out bp
Symmetric	informed	normal	4843	0.45	+0.14
Symmetric	informed	pre-event	282	0.45	+1.99
Symmetric	real_money	normal	9630	0.54	+0.04
Symmetric	real_money	pre-event	564	0.54	+0.06
Symmetric	other	normal	14390	0.57	-0.00
Symmetric	other	pre-event	294	0.57	+0.21
Bergault	informed	normal	4956	0.41	+0.13
Bergault	informed	pre-event	293	0.44	+1.93
Bergault	real_money	normal	9167	0.54	+0.05

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Bergault	real_money	pre-event	541	0.54	+0.03
Bergault	other	normal	12723	0.62	-0.01
Bergault	other	pre-event	252	0.60	-0.02
BarzykinCiceri_QAHR	informed	normal	3617	0.53	+0.15
BarzykinCiceri_QAHR	informed	pre-event	209	0.54	+2.15
BarzykinCiceri_QAHR	real_money	normal	7487	0.63	+0.05
BarzykinCiceri_QAHR	real_money	pre-event	474	0.59	+0.02
BarzykinCiceri_QAHR	other	normal	11547	0.67	-0.01
BarzykinCiceri_QAHR	other	pre-event	243	0.65	-0.07
CarteaWang	informed	normal	4954	0.41	+0.14
CarteaWang	informed	pre-event	50	0.96	+1.73
CarteaWang	real_money	normal	9181	0.54	+0.05
CarteaWang	real_money	pre-event	415	0.66	+0.07
CarteaWang	other	normal	12726	0.62	-0.00
CarteaWang	other	pre-event	205	0.73	-0.10
Bergault_oracle	informed	normal	3176	0.55	+0.06
Bergault_oracle	informed	pre-event	0	0.00	+0.00
Bergault_oracle	real_money	normal	8364	0.58	+0.05
Bergault_oracle	real_money	pre-event	81	1.26	-0.01
Bergault_oracle	other	normal	12742	0.62	-0.02
Bergault_oracle	other	pre-event	253	0.61	-0.02

Real-day replay, factor model fitted without 2020-22: 2024-01-01..2026-12-31, 676 days, 186 event days (21 w)

Events by type: AUCTION 111, ECB 22, FOMC 22, NFP 31.

										quoter
Symmetric	text+calendar	153.7	[148.2, 157.0]	20.1	0.38	185.4	-9.2	-22.4	0.35/0.35/0.35	
Bergault	text+calendar	147.5	[144.9, 150.0]	12.1	1.01	166.5	-5.0	-14.1	0.34/0.32/0.30	
BarzykinCiceri_QAHR	text+calendar	144.4	[142.9, 146.4]	12.0	1.01	163.6	-4.8	-14.4	0.25/0.26/0.27	
CarteaWang	text+calendar	149.2	[146.1, 152.5]	12.0	1.04	165.3	-2.3	-13.9	0.33/0.32/0.30	
CarteaWang	placebo_text+calendar	149.7	[145.7, 154.2]	12.0	1.04	165.2	-1.7	-13.8	0.33/0.32/0.30	
CarteaWang	calendar_only	149.3	[146.2, 153.3]	12.0	1.04	165.4	-2.3	-13.9	0.33/0.32/0.30	
CarteaWang	none	147.5	[144.9, 150.0]	12.1	1.01	166.5	-5.0	-14.1	0.34/0.32/0.30	
Bergault_oracle	text+calendar	142.3	[138.8, 144.4]	11.5	1.08	154.0	-0.0	-11.6	0.21/0.28/0.30	

quoter	tier	window	fills	half-spread bp	30m mark-out bp
Symmetric	informed	normal	4843	0.45	+0.10
Symmetric	informed	pre-event	282	0.45	+2.14
Symmetric	real_money	normal	9630	0.54	+0.01
Symmetric	real_money	pre-event	564	0.54	-0.01
Symmetric	other	normal	14390	0.57	-0.02
Symmetric	other	pre-event	294	0.57	+0.34
Bergault	informed	normal	4809	0.39	+0.09
Bergault	informed	pre-event	260	0.42	+2.38
Bergault	real_money	normal	8930	0.52	+0.05
Bergault	real_money	pre-event	526	0.53	-0.18
Bergault	other	normal	12455	0.61	+0.01
Bergault	other	pre-event	253	0.59	+0.19
BarzykinCiceri_QAHR	informed	normal	3512	0.50	+0.16
BarzykinCiceri_QAHR	informed	pre-event	212	0.50	+2.66
BarzykinCiceri_QAHR	real_money	normal	7232	0.62	+0.04
BarzykinCiceri_QAHR	real_money	pre-event	453	0.58	+0.02
BarzykinCiceri_QAHR	other	normal	11073	0.67	-0.02
BarzykinCiceri_QAHR	other	pre-event	229	0.66	+0.17
CarteaWang	informed	normal	4806	0.39	+0.08
CarteaWang	informed	pre-event	51	0.88	+3.26
CarteaWang	real_money	normal	8957	0.52	+0.08
CarteaWang	real_money	pre-event	399	0.65	+0.03
CarteaWang	other	normal	12464	0.60	+0.01

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CarteaWang	other	pre-event	208	0.73	+0.16
Bergault_oracle	informed	normal	3155	0.52	+0.03
Bergault_oracle	informed	pre-event	0	0.00	+0.00
Bergault_oracle	real_money	normal	8142	0.56	+0.08
Bergault_oracle	real_money	pre-event	15	1.62	+1.49
Bergault_oracle	other	normal	12447	0.61	-0.00
Bergault_oracle	other	pre-event	258	0.60	+0.08

Tests

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